ABU Robocon 2027 — Competition Strategy Manual

"The Pursuit of Mustika Nusantara" | Solo, Indonesia

SECTION 1: STRATEGY MANUAL

1. Overall Game Objective

- Match format: Red vs Blue, exactly 3 minutes (180 s), no early win — highest total score at the buzzer wins.
- Two-robot system:
- Transporter Robot (TR): manual or automatic; operates ONLY in the Ground Area + Transfer Area; harvests Earth Blocks (Storage Area), Sky Blocks (Ground Shared Area) and the Mustika; delivers to the Transfer Area (5 pts per block delivered).
- Builder Robot (BR): FULLY AUTONOMOUS; operates in Transfer Area, Level 1 (L1) and Level 2 (L2); builds Complete Towers (Earth Block -> Earth Block -> Sky Block, correct team color up) on 500x500 mm Building Spots; enshrines the Mustika on the Central Pillar (L2).
- Point economy (per block placed on a Building Spot):

Object	L1	L2
Earth Block (bottom)	10	20
Earth Block (middle)	20	40
Sky Block (top)	40	80
Complete Tower total	70	140
Mustika enshrinement	—	250

- Core strategy principle: secure stable base points (L1 towers + transfer points) FIRST, unlock the 250-point Mustika mission as early as possible, then build L2 towers. Offensive Sky-Block flipping is OPTIONAL — only when ahead on schedule.

2. Phase-Based Match Strategy (180 s)

Phase 1 — Early (0–90 s): Base Points + Sanctuary Mandate

- TR: shuttle Earth Blocks from Storage Area to Transfer Area at max payload (3 blocks/trip = 15 transfer pts/trip). Also fetch 2 Sky Blocks from the Ground Shared Area.
- BR: climb to L1 via the RAMP (3500 mm, far safer than 150 mm stairs) immediately at start — this also unlocks eligibility for the L1 Retry Zone later.
- BR builds 2 Complete Towers on L1: Tower A on our exclusive L1 spot, Tower B in the L1 Shared Area.
- Material requirement for Phase 1: 4 Earth Blocks + 2 Sky Blocks = 6 blocks = exactly 2 full TR trips.
- Milestone: Sanctuary Mandate achieved (>= 2 Complete Towers, >= 1 in a shared zone). Once achieved it is PERMANENT — opponents cannot take it away by flipping our towers.
- Target score by 90 s: 2 L1 towers (140 pts) + transfer points (~30 pts) = ~170 pts.

Phase 2 — Mid (90–150 s): Mustika Mission (250 pts)

- TR (now free on the ground) drives to the Mustika Pillar, retrieves the Mustika (volleyball, ~270 g, concave pedestal top), delivers it to the Transfer Area. NOTE: while touching the Ground Area, TR cannot release objects into the Transfer Area — TR must drive up into the Transfer Area boundary before release.
- BR receives the Mustika, climbs L1->L2 stairs, enshrines it on the Central Pillar (800 mm high, concave receptacle 180 mm dia x 100 mm deep).
- Must be placed STABLE and RELEASED — anything still touched/gripped at the buzzer scores 0.
- If completed early: BR begins L2 tower construction (140 pts each) with remaining Earth/Sky Blocks.

Phase 3 — Late (150–180 s): Lock-In + Optional Offense

- Hard rule: at T-10 s, ALL blocks and the Mustika must be fully released. Nothing held counts.
- BR finishes current placement early rather than risking a held block at the buzzer.
- Optional only if time permits: flip opponent Sky Blocks in shared zones (steals 40/80 pts per tower). Never sacrifice a release deadline for a flip.
- TR parks clear of the Transfer Area to avoid accidental contact with scored blocks.

3. Robot Hardware — Key Components

Transporter Robot (TR) — recommended: manually operated (rules allow it; far lower risk than full auto)

- Chassis: 4WD omnidirectional (mecanum) base, start size <= 700x700x700 mm; extended <= 1000x1400x1200 mm; weight <= 50 kg.
- Block gripper: clamp or suction (suction explicitly allowed; Sky Blocks have reinforced claws for suction) sized for both 350 mm Earth Blocks (~600 g) and 200 mm Sky Blocks (~220 g); capacity 3

blocks (stack-and-carry magazine design).

- Mustika cradle: dedicated scoop/claw for a 210 mm sphere; secure retention over ramp slopes.
- Operator link: Wi-Fi/Zigbee/Bluetooth ONLY (802.11/802.15/BT); plan for congested venue RF — organizer does NOT guarantee signal quality. Wired backup controller strongly advised if permitted.
- FPV camera + LEDs for operator depth judgment at the Transfer Area handover.
- Red emergency STOP button (mandatory, visible + accessible); remote E-stop encouraged.

Builder Robot (BR) — fully autonomous

- Chassis: holonomic base capable of 150 mm stair climbing OR reliable ramp ascent; ground clearance for the L1->L2 stairs (300 mm run x 150 mm rise steps).
- Lift/stack mechanism: vertical gantry or 4-bar lift reaching tower height: 2x Earth Blocks (700 mm) + placement margin; Mustika enshrinement needs reach to 800 mm Central Pillar top + receptacle engagement.
- Dual block carriage: capacity 2 blocks (rule limit) with independent release order (Earth first, Sky last).
- Sky Block flip tool (optional module): small top-side gripper to rotate a 200 mm block 180 deg in place — deploy only if Phase 1-2 are consistently reliable in testing.
- Alignment aids: downward camera or laser dot for Building Spot (green 500x500 mm) centering; ToF/ultrasonic rangers for tower vertical alignment; load-switch confirmation of successful release.
- Red emergency STOP button (mandatory).

4. Positioning & Sensor Solution (BR)

Because TR and BR are forbidden to communicate, BR must self-localize for the entire match:

- Primary localization: 2D LiDAR (Class 1 laser only, IEC 60825-1) + wheel odometry + IMU, fused with an EKF; field is walled/structured (fences, barriers, pillars) — good LiDAR features.
- Fallback/augment: floor color sensing (field zones have distinct RGB-specified paint) + tape/mark line following along known waypoints; encoder-only dead reckoning between LiDAR dropouts.
- Zone-boundary guarding: infrared/ToF edge sensors facing the Center Divider and L1 Perimeter Barrier; software geofence with hard motor cut before vertical projection crosses any restricted boundary (zone violations = forced retry).
- Cliff/edge detection on L1/L2 outer edges despite perimeter barriers.
- Block detection at Transfer Area: short-range ToF/ultrasonic grid + camera (optional) — remember BR may only touch objects fully inside the Transfer Area (including its airspace).
- TR (manual): no autonomy sensors required; add wheel encoders + buzzer timer display for the operator.

5. Critical Rule Constraints (design around these)

- NO TR<->BR communication (Rule 11.10): handover must be passive — fixed floor drop points inside the Transfer Area, pre-programmed BR pickup sequence, or optical markers BR can read. No radio, no sync signals.
- Payload limits: TR max 3 blocks, BR max 2 blocks at any time.
- Zone boundaries judged by VERTICAL PROJECTION of ANY part (incl. mechanisms): TR never enters L1 beyond Transfer Area or L2; BR stays off the Ground Area except stairs/retries; neither enters opponent territory except shared zones.
- Sanctuary Mandate is a PRE-CONDITION: TR may NOT touch the Mustika before 2 Complete Towers (>= 1 in shared zone) exist. Sequence software must hard-block early retrieval.
- No pushing/dragging blocks — everything must be actively lifted/carried (incl. dropped blocks).
- Handover violation if BR touches objects not completely inside the Transfer Area, or TR releases while still touching Ground Area.
- Anything touched/gripped/supported at the buzzer = 0 points. Release discipline is scoring discipline.
- Setup: both robots fully inside 700x700 mm Start Zones (nothing overhanging), 20 Earth Blocks stacked max 2-high in Storage Area, 1-minute setup.
- Technical limits: battery <= 24 V nominal (circuits <= 42 V), <= 50 kg per robot, compressed air <= 600 kPa, no drones/flying mechanisms, no lead-acid batteries.

6. Retry Usage

- Retries are unlimited and self-requested ("Retry!" -> referee acknowledges). Use them EARLY and decisively — a stuck/mislocalized robot costs more time than a clean reset.
- BR retry choice: use the L1 Retry Zone whenever possible (700x700 mm on our L1 side) — but BR must have COMPLETELY entered L1 at least once first (another reason to climb the ramp in the first 10 seconds).
- Held blocks are KEPT on retry (except the Mustika, which always returns to its pillar) — so a retry does not waste loaded cargo.
- NEVER retry while carrying the Mustika unless the robot is truly dead — it goes all the way back to the Mustika Pillar and the retrieval must be redone.
- Violations (pushing, zone crossing, handover faults, shared-area interference) trigger FORCED retries — design geofences and release checks to make violations structurally impossible.
- During opponent retries: keep clear — interfering with their retry is itself a violation.

7. Risk-Mitigation Rules (team checklist)

1. Stability first: never start an L2 tower or a flip while the Mustika mission is incomplete.
2. Every placement ends with a sensor-confirmed release (load switch / current drop) before moving on.
3. Software geofence on both robots; mechanical bumpers as last-resort protection at fences.
4. Drop detection: if a block is dropped, only re-grasp by lifting; never nudge it into place.
5. Avoid the Ground Shared Area when the opponent is inside it — interference rulings go against the initiator of contact.
6. RF contingency: pre-scan venue Wi-Fi channels at test-run day; keep manual TR controls on the cleanest link; fail-safe = stop in place, never keep driving blind.
7. Endgame discipline: T-15 s all mechanisms to neutral, all cargo released; operator verbal countdown.
8. Pre-match checks: E-stop function, battery voltage, block stack <= 2 high in Storage Area, robot fully inside Start Zone, score-sheet review BEFORE signing (signed = final).
